Lab: Implementing Data Modeling in Tableau

Estimated time needed: 15 minutes

In this lab, you will learn how to implement data modeling in Tableau.

Required Software

You can complete this lab using the free downloadable **Tableau Desktop (Public Edition)** desktop software. You may also use the free **Tableau Public online platform**, but some tasks may not work in exactly the same way or may not be supported in that version.

Prerequisites

You need a New Tableau Public account for this lab. If you haven't already done so, you can sign up for an account using the steps here.

Disclaimer: The screenshots in this lab are taken from Tableau Desktop. If you are using Tableau Public, you may notice some differences due to version variations.

Data Set

This lab uses several sample data sets called **Country.csv**, **Indicators_small.csv**, and **Series.csv** to showcase the Tableau features used in these lab exercises.

Install the CTRL key while you click to download each of the following files:

Country.csv

Indicators_small.csv

Series.csv

Objectives

The exercises in this lab teach you how to implement data modeling in Tableau.

In this hands-on lab, you will:

1. Implement data modeling.

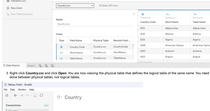
Exercise: Implement Data Modeling

In this exercise, you will learn how to join tables by joining the **Country** table with the **Indicators_small** table.

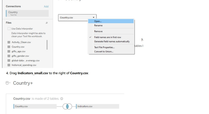
1. Under **To a File**, click **Text file**.

2. Navigate to the folder where you downloaded the data sets to, then select **Country.csv**, and click **Open**.

3. Right-click **Country.csv** and click **Open**. You are now viewing the physical table that defines the logical table of the same name. You need to view the physical table as you are done between physical tables, not logical tables.



4. Drag **Indicators_small.csv** to the right of **Country.csv**.



5. Click the **Inner Join** symbol.

6. You can see these two tables are now part of an Inner Join, and you can see the four types of join (Inner, Left, Right, and Full Outer). The common field in this join is Country Code.

7. Close the **Join** window.

8. You'll now join the Indicators_small table with the Series table, by dragging **Series.csv** to the right of **Indicators_small.csv**.

9. Click the **Inner Join** symbol.

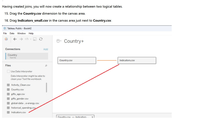
10. You can see these two tables are now part of another Inner Join, and you can see that the common field in this join is Indicator Name.

11. Close the **Join** window and click **X** in the top right corner of the canvas.

12. Hover over the **Country.csv** data file.

13. You can see that Country.csv is one logical table combined with three physical tables: Country.csv, Indicators_small.csv, and Series.csv.

14. Click the little arrow to the right of **Country.csv** and click Remove.



Having created joins, you will now create a relationship between two logical tables.

15. Drag the **Country.csv** dimension to the canvas area.

16. Drag **Indicators_small.csv** to the canvas area just next to **Country.csv**.

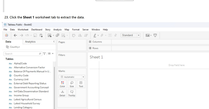
17. Hover over the orange line (the **Inactive** joining **Country.csv** and **Indicators_small.csv**.

18. You can see that the relationship type (or Cardinality) is the default Many to Many relationship, and the common (or Related field is Country Code.

19. In the bottom left of the canvas, expand **Performance Options** > **Cardinality**, select **One** from the first drop-down list.

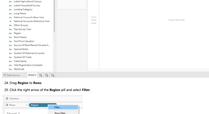
20. Hover over the results joining **Country.csv** and **Indicators_small.csv**.

21. You can see that this relationship is now a One to many relationship.

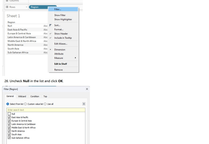


Now that you've created a join and a relationship between two or more tables, next, you'll visualize the data for presentation.

22. In the data window, click **Update Now**.



23. Click the **Sheet 1** worksheet tab to extract the data.



24. Drag **Region** to **Rows**.

25. Click the right arrow of the **Region** pill and select **Filter**.

Tableau Prep - Book1
File Connect Dashboard Data Analytics Map Format Sort Window Help
Connections
Country
Country.csv
Tables
Sheet 1
Sheet 1
Region
Drop Fields Here
Filter
Remove
Rename
Duplicate
Sort...
Hide
Edit Alias...
Edit Calculation...
Edit Hierarchy...
Create Hierarchy
Add to Hierarchy
Remove from Hierarchy
Move to Top
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